Submission Summary

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Paper Title
BRAIN BARTER: A PLATFORM FOR PEER-TO-PEER SKILL EXCHANGE

Abstract
This paper presents Brain Barter, a full-stack web application designed to facilitate monetary-free skill exchanges within communities through intelligent matching algorithms and real-time communication infrastructure. The platform addresses critical barriers in traditional education systems, including high costs, geographical limitations, and inadequate peer-to-peer learning mechanisms. Built on a modern technology stack comprising React.js, Node.js with Express, MongoDB, and Socket.IO, the system implements a three-tier architecture with robust authentication, real-time messaging, and a credit-based exchange mechanism. Performance analysis demonstrates average matching times of 2.3 seconds and a 78% successful match rate within three suggestions. The implementation includes comprehensive security measures, including JWT-based authentication, rate limiting, and input sanitization. User acceptance testing with 50 participants indicates 92% satisfaction with the matching system and 88% appreciation for the elimination of monetary transactions. This work contributes to the growing field of collaborative learning platforms by providing a scalable, secure, and user-centric solution for democratizing knowledge sharing.

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Submission Files

Brain-Barter.pdf (703.6 Kb, 20/02/2026, 19:17:28)